

Milestone 2.5: Synthesis: One Cohort, A Whole Semester

INFO 521 · Machine Learning Foundations · project milestone brief ·
From Inference to Discovery · Synthesis

How to read this

This is the assignment brief for one project milestone. The Satisfactory criteria below are authoritative: every one must be met, and it is all-or-nothing rather than partial credit. A Not-Yet returns with targeted feedback and one revise cycle. Do the work in the notebook distributed through Classroom 50; this PDF is the brief, not the workspace.

Course-schedule placement. Part 2 of the project: begins at the midpoint, due Week 7.5. This milestone covers M5 to M7 and is due Week 7.5. Gated by structured peer review (synthesis).

The whole arc

One cohort has carried every method this semester. You met it as a regression problem (*how does systolic BP change with age?*) and answered it three ways (least squares, maximum likelihood, Bayesian). Part 2 turned the **same patients** into a classification problem (*who is hypertensive?*), answered it two ways (probabilistic vs. margin), then dropped the labels entirely and asked whether the structure was there all along. This synthesis makes that arc a single, argued story.

Ground rules. Reuse **your own** from-scratch implementations from 1.1–2.4: import your modules or copy your functions. No scikit-learn or off-the-shelf solvers, here or anywhere upstream. info521 stays plumbing-only.

What “Satisfactory” means for 2.5

- ☐ You tell **one narrative across the whole semester on the same cohort**: P1 regression (age → SBP, three lenses) → Gate A supervised hypertension classification (probabilistic vs. margin) → Gate B unsupervised structure (does it recover the strata?).
- ☐ You **argue supervised vs. unsupervised framing**: what question each mode answers, how the methods relate, and which you would reach for given a **new** clinical question.
- ☐ The notebook **runs end-to-end** against the pinned environment.
- ☐ Structured **peer review** completed (rubric posted on D2L).

Setup

```
import numpy as np
from info521 import data, plotting

plotting.use_course_style()
```

```

ds = data.load_clinical()

age = data.primary_predictor(ds)      # P1's single predictor
X = ds["X"]                          # Gate A/B six-feature space (non-BP)
sbp = ds["y"]                        # P1 continuous target
labels, rule = data.hypertension(ds)  # Gate A label / Gate B reveal
print(f"{X.shape[0]} patients · features {ds['features']}")
print(f"P1 target = {ds['target']} (SBP) · P2 label = {rule}")

```

1 · The arc in one figure

Assemble a **single multi-panel figure** that walks the semester on this one cohort. A clean three-panel layout:

- **Panel A: Part 1 (regression).** age → SBP with your best fit and a posterior-predictive band (the three-lens result from 1.4).
- **Panel B: Gate A (supervised).** the hypertension decision viewed on two features (your Bayesian-logistic probability surface and the SVM margin on the same axes) with precision/recall/F1 reported.
- **Panel C: Gate B (unsupervised).** the PCA projection from 2.4 colored by your k-means clusters, with the hypertension label alongside.

```

# TODO: import/copy your fitted models from 1.4, 2.1/2.2, 2.3/2.4 and draw the
# three panels. This is integration, not re-derivation: reuse your own code.

```

2 · The argument

Address, in prose (≈500–800 words total):

1. **Supervised vs. unsupervised.** What question does each *mode* answer? Gate A was handed a target and asked to predict it; Gate B was handed nothing and asked to find structure. State precisely what each can and cannot tell a clinician.
2. **How the methods relate.** Trace the throughline: the conjugate Gaussian posterior of 1.3 → its deliberate break in 2.1 (approximate inference) → the discriminative reframing in 2.2 (margin, not probability) → structure without labels in 2.3–2.4. Where does each method sit on the axes of *probabilistic* ↔ *geometric* and *supervised* ↔ *unsupervised*?
3. **Recovery.** Did the unsupervised subgroups recover the hypertension strata (your 2.3 cross-tab and 2.4 projection)? What does the answer (full, partial, or none) say about whether clinical structure is *geometric* structure?
4. **Choice.** Given a **new** clinical question on this cohort (you state it), which lens/mode would you reach for first, and why?

3 · Reflection (1 paragraph)

Across the semester you saw the *same* prediction as a point, a likelihood, a posterior; then as a probability vs. a margin; then as discovered structure. What did moving across all of these teach you that no single method, and no single project, could have?

i Gate: structured peer review

Per the course peer-review rubric (posted on D2L), the Part 2 synthesis (this milestone) carries a **structured, rubric-guided peer review** (the supervised-gate play artifact carries the lighter mid-stream review). Reviewer feedback feeds your revise cycle; it does not assign the Satisfactory/Not-Yet judgment. Not Satisfactory until the review is completed.

Looking back

P1 asked *how much?* and answered with an estimate and its uncertainty. P2 asked *which kind?* and *what structure?*, and answered with boundaries and with clusters. Same patients, three questions, one toolkit you built from scratch.